

Experiment 2.1: Design and Performance Evaluation of a Chebyshev Type-I IIR Low Pass Filter

MATLAB Code:

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%% Experiment 2.1: Design and Performance Evaluation of a Chebyshev Type-I IIR Low Pass Filter

clc;
clear;
close all;

%% Specifications
Fs = 1000;      % Sampling Frequency (Hz)
Fp = 100;       % Passband Frequency (Hz)
Fs1 = 200;      % Stopband Frequency (Hz)

Rp = 1;         % Passband Ripple (dB)
Rs = 40;        % Stopband Attenuation (dB)

%% Normalize Frequencies
Wp = Fp/(Fs/2);
Ws = Fs1/(Fs/2);

%% Find Filter Order
[n, Wn] = cheb1ord(Wp, Ws, Rp, Rs);

fprintf('Filter Order = %d\n', n);
fprintf('Cutoff Frequency = %.3f\n', Wn);

%% Design Filter
[b, a] = cheby1(n, Rp, Wn, 'low');

%% Compute Responses
[H, f] = freqz(b, a, 512, Fs);
[h_imp, t_imp] = impz(b, a);
[h_step, t_step] = stepz(b, a);
[gd, w_gd] = grpdelay(b, a, 512, Fs);
[z, p, k] = tf2zp(b, a);

%% Plot Outputs
figure;
subplot(3,2,1);
plot(f, 20*log10(abs(H)));
title('Magnitude Response (dB)');

subplot(3,2,2);
plot(f, angle(H));
title('Phase Response');

subplot(3,2,3);
stem(t_imp, h_imp);
title('Impulse Response');

subplot(3,2,4);
stem(t_step, h_step);
title('Step Response');

subplot(3,2,5);
plot(w_gd, gd);
title('Group Delay');

subplot(3,2,6);
zplane(b, a);
title('Pole-Zero Plot');

exportgraphics(gcf, 'exp2_1_full_output.png', 'Resolution', 300);
```

Output:

